ChatGPT

## Research Results Summary

The structured literature search identified a substantial number of relevant papers in each query category (Q1–Q6). The table below summarizes the counts of **total relevant papers**, those published **2022–present**, and the subset of **"Mature SoA" papers** (i.e. peer-reviewed works with citations or high relevance) for each category:

| Query Category | Total Papers | 2022–Present | Mature SoA Papers |
|---|---|---|---|
| **Q1: Classical / Observers** | 11 | 10 | 9 |
| **Q2: Gaussian Process (GP) / Hybrid GP** | 3 | 3 | 2 |
| **Q3: Deep Sequence Models (MLP/GRU/...)** | 7 | 5 | 4 |
| **Q4: Hybrid / Residual** | 4 | 3 | 3 |
| **Q5: Physics-Informed / Differentiable** | 2 | 2 | 2 |
| **Q6: Domain Adaptation / Latent Context** | 9 | 6 | 2 |

*(Note: "Total Papers" excludes non-article references like textbooks. "2022–Present" counts publications from 2022 to 2025. "Mature SoA Papers" are a selected subset of highly relevant works in 2022–2025, often in top venues or showing notable impact.)*

## Analysis of Mature SoA Papers (2022–2025)

Below we analyze each **Mature State-of-the-Art paper** (2022–2025), describing its approach and identifying whether its focus is on **rigid-body estimation**, **payload estimation**, or **both**:

- **Nadeau et al. (ICRA 2022)** – *"Fast Object Inertial Parameter Identification for Collaborative Robots."* This work targets **payload estimation**. It addresses the low signal-to-noise ratios (SNR) in collaborative robot motion by using an approximate rigid-body dynamics model to boost SNR, and a point-mass discretization technique to incorporate object shape information [1]. The method enables faster and more robust identification of a manipulated object's inertial parameters (mass, center of mass, inertia) in the cobot's operating regime [2]. *Focus: Payload estimation.*

- **Nadeau et al. (ICRA 2023)** – *"The Sum of Its Parts: Visual Part Segmentation for Inertial Parameter Identification of Manipulated Objects."* This paper also focuses on **payload estimation**. It combines **vision** (RGB-D based part segmentation) with force-torque sensing to perform safe inertial identification of unknown objects [3]. By segmenting an object into homogeneous parts (HPS method) and then estimating each part's inertial parameters, the approach requires only slow, stop-and-go motions (improving safety around humans) yet achieves accurate full inertial parameter identification [4] [5]. The authors validate the approach on a dataset of tools and a real "hammer balancing" demo, showing it outperforms traditional fast-motion identification methods. *Focus: Payload estimation.*

- **Kurdas et al. (ICRA 2022)** – *"Online Payload Identification for Tactile Robots Using the Momentum Observer."* This study is on **payload estimation** in real time. It integrates a **momentum-based observer** (to estimate external torques) with a recursive least-squares scheme and a filtering/calibration strategy for online payload identification [6]. Tested on a Franka Panda robot, the method could identify changes in payload (mass) on the fly, achieving lower estimation error than baseline methods while using only the robot's joint torques and kinematic data [6]. *Focus: Payload estimation.*

- **Kommuri et al. (IEEE/ASME T-Mech 2022)** – *"External Torque Estimation Using Higher Order Sliding-Mode Observer for Robot Manipulators."* This paper addresses **rigid-body external force estimation**. It designs a **high-order sliding-mode (HOSM) observer** to estimate external torques/forces on a manipulator in real time [7]. The observer is robust against nonlinear friction and model uncertainties, and is augmented by a Luenberger observer (via sum-of-squares tuning) to stabilize the momentum error dynamics [8]. Lyapunov stability proofs are provided. Experiments on a 7-DOF Sawyer robot demonstrate that the scheme can accurately track time-varying external torques under various scenarios, outperforming an extended state observer baseline [9]. *Focus: Rigid-body (external force) estimation.*

- **Cao et al. (Robotics & CIM 2021)** – *"Contact force and torque sensing for a serial manipulator based on an adaptive Kalman filter."* (Pre-2022, but a notable mature reference.) This work focuses on **rigid-body force estimation**. It uses an **adaptive Kalman filter** with variable update rate to estimate contact forces/torques on a robot using only motor currents [10] [11]. The filter adapts to model uncertainty and noise (using a mode-switching moving average to adjust covariances), allowing sensorless force estimation. Tests on a UR5 arm showed reduced force error and robustness to payload changes [12] [13]. *Focus: Rigid-body (external force) estimation.*

- **Zhang et al. (EECR 2025)** – *"Accurate Payload Dynamics Estimation and Compensation of a Robotic Manipulator without External Motion Measuring Sensors."* This recent conference paper deals with **payload estimation**. It proposes an online identification method that requires no external motion sensors (no IMU, no optical tracker), using only a wrist force/torque sensor and the robot's own encoders [14]. The approach decouples the identification problem into steps (first identifying sensor biases, then payload mass & CoM, then inertia tensor) and employs optimized Fourier-trajectory excitation and second-order filtering [15]. In experiments on a 7-DoF manipulator, it achieved <10% error in payload mass and significantly improved the compensation of payload-induced forces/torques during robot motions [16] [17]. *Focus: Payload estimation.*

- **Hu et al. (IEEE T-RO 2025)** – *"On the Fully Decoupled Rigid-Body Dynamics Identification of Serial Industrial Robots."* This journal article addresses **both** robot and payload identification. The authors present a **fully decoupled identification framework** that separates the estimation of the robot's own dynamics and the payload's dynamics [18]. By designing specialized excitation trajectories (reciprocating S-curve motions) that individually excite friction, link inertias, and payload parameters, they achieve independent estimation of each component's parameters [19]. The method improved torque prediction accuracy and payload mass/CoM estimation compared to prior coupled identification methods (CRDI/PDRDI) [20]. This is the first work enabling truly independent identification of robot and payload dynamics in one system. *Focus: Both (rigid-body and payload).*

- **Liu et al. (IEEE T-ASE 2025)** – *"A Two-Stage Payload Dynamic Parameter Identification Method for Interactive Industrial Robots with Large Components."* This work focuses on **payload estimation** for heavy payloads. It introduces a two-stage identification: a **static identification** to get initial payload parameters, followed by a **dynamic identification** using a **recursive restricted total least squares (RRTLS)** estimator to refine the parameters under motion [21]. The authors also plan an optimal excitation trajectory within the robot's workspace to minimize the impact of modeling errors [22]. On a 35 kg payload (aircraft component), the method achieved very low force/torque estimation errors (<3 N and <1.7 Nm) and outperformed standard LS and TLS approaches [23], demonstrating its effectiveness for large payloads [24]. *Focus: Payload estimation.*

- **Xu et al. (IEEE TIM 2025)** – *"Identifying Current Dynamics of Robot Payload Based on Iterative Weighting Estimation."* This paper proposes a novel **payload estimation** approach that relies on **motor current signals** rather than torque sensors [25]. By modeling the relationship between motor currents and payload dynamics (including friction), the authors perform an **iterative weighted least squares** identification of payload parameters (a total of 56 parameters, incorporating payload inertia and nonlinear friction) [26]. This current-based method avoids uncertainties in torque sensor calibration. Experiments on a UR10 robot showed it achieved the lowest payload estimation errors compared to four torque-based identification methods, and it improved the robot's collision detection sensitivity when the payload changes [27]. *Focus: Payload estimation.*

- **Xu et al. (Robotica 2022)** – *"An accurate identification method based on double weighting for inertial parameters of robot payloads."* This journal article centers on **payload estimation**. It observes that prior payload ID methods didn't fully exploit data weighting to reduce outlier effects [28]. The authors introduce a **double-weighting** framework: first performing a two-loop dynamic identification to estimate the robot's base dynamics and obtain preliminary weights, then using those weights in a dedicated payload parameter estimator [29]. Their method explicitly accounts for payload-induced changes in friction as well [30]. In comparative experiments, the double-weighting approach achieved the highest accuracy, particularly for payload mass and center-of-mass estimates, improving errors by an order of magnitude (10–43×) over standard least-squares methods [31]. *Focus: Payload estimation.*

- **Duan et al. (MDPI Sensors 2022)** – *"Payload Identification and Gravity/Inertial Compensation for 6-D F/T Sensor with a Fast and Robust Trajectory Design."* This paper addresses **payload estimation**, with an eye on improving force/torque sensor readings. It uses a **trajectory optimization approach** to robustly identify payload inertial parameters and then compensate for the payload's weight and inertial forces in a 6-axis force/torque sensor [32]. By designing an excitation trajectory that minimizes the coupling between payload dynamics and measurement noise, the method achieves fast convergence of the parameter estimates and improves the accuracy of force/torque measurements under payload influence [33]. The result is a more reliable contact force sensing during robot operation with a payload, as verified experimentally. *Focus: Payload estimation.*

- **Wei et al. (IEEE T-ASE 2024)** – *"Contact Force Estimation of Robot Manipulators with Imperfect Dynamic Model: On Gaussian Process Adaptive Disturbance Kalman Filter."* This work focuses on **rigid-body force estimation** under model uncertainties. It proposes a **Gaussian Process Adaptive Disturbance**

**Kalman Filter (GP-ADKF)** that augments a Kalman filter with a Gaussian process model to learn unmodeled dynamics (e.g., friction, disturbances) [34]. The GP provides a probabilistic estimate of the residual dynamics which is fed into the Kalman filter as an adaptive disturbance term [35]. An online variational Bayesian scheme adjusts the noise covariance in the Kalman filter. Tests showed this GP-augmented observer outperforms standard disturbance Kalman filters and extended disturbance observers, yielding more accurate contact force estimates in the face of modeling errors [36] [37]. *Focus: Rigid-body (contact force) estimation.*

- **Wei et al. (IEEE CCIS 2022)** – *"Decoupling Observer for Contact Force Estimation of Robot Manipulators Based on Enhanced Gaussian Process Model."* This conference paper (a precursor to the above) also deals with **rigid-body force estimation**. It combines a **momentum-based decoupling observer** (to isolate interaction forces) with an **enhanced Gaussian process regression** model that learns the robot's unmodeled dynamics [37]. The GP model improves the observer's ability to estimate contact forces in the presence of uncertainties (e.g., unknown payloads or friction). Experiments on a 3-DOF robot arm showed the GP-enhanced observer was more robust and accurate than a standard disturbance KF or a neural-network-based observer, especially when the robot's dynamics were imperfect [38] [39]. *Focus: Rigid-body (contact force) estimation.*

- **Fathi et al. (ICMERR 2022)** – *"Human-Robot Contact Detection in Assembly Tasks (using GP classifier)."* This is a slightly different angle on **rigid-body interaction** estimation – focusing on contact *detection* rather than force magnitude. The authors use a **Gaussian process classifier** to probabilistically detect whether a contact has occurred during a robot assembly task [40]. The GP model takes in torque sensor readings and possibly other signals, and outputs a probability of contact vs no-contact, with uncertainty estimates. The GP-based approach was found to be more robust to noise and uncertainties than simple threshold methods or deterministic classifiers [41]. This contributed to safer human-robot collaboration by reducing false positives/negatives in collision detection. *Focus: Rigid-body (external contact) estimation.*

- **Lao et al. (IEEE RA-L 2023)** – *"A Learning-Based Approach for Estimating Inertial Properties of Unknown Objects from Encoder Discrepancies."* This article focuses on **payload estimation** using deep learning. The authors present a method to identify an **unknown payload's mass and center-of-mass** using only the robot's **encoder data** (joint angle readings) [42]. They derive that when a manipulator carries an unknown payload, there will be small discrepancies between expected joint motions (from a nominal model) and actual motions; a neural network (CNN) is trained on these discrepancy signals to predict payload mass and CoM [43]. Furthermore, an **attention mechanism** weights the contribution of each joint's signal, improving estimation accuracy [44] [45]. This encoder-only approach was able to estimate payload inertia properties with high accuracy, offering a practical alternative when force sensors are unavailable. *Focus: Payload estimation.*

- **Kružić et al. (Electronics 2021)** – *"End-Effector Force and Joint Torque Estimation of a 7-DoF Manipulator Using Deep Learning."* (Included as a key background study.) This work dealt with **rigid-body force/ torque estimation** and compared various deep sequence models. It found that a recurrent neural network (LSTM) best captured the temporal patterns of the system, outperforming feed-forward networks in estimating the robot's end-effector forces and joint torques from proprioceptive data [46] [47]. The insight is that sequence models (LSTM) can learn the dynamics better and filter noise over time, highlighting the importance of temporal modeling for accurate sensorless force estimation. *Focus: Rigid-body (force/torque) estimation.*

- **Pan et al. (Eng. Apps of AI 2023)** – *"An adaptive sparse general regression neural network-based force observer for teleoperation."* This paper is about **rigid-body force estimation** under a teleoperation context. The authors use a **general regression neural network (GRNN)** – a memory-based non-parametric model – and make it **sparse** and adaptive [48]. They introduce feature selection and an ant-lion optimizer to prune the network and tune its kernel bandwidth. The resulting force observer can estimate contact forces on a slave manipulator without force sensors. It achieved over an *80% reduction in MSE* compared to Gaussian process, MI-NN, or random forest baselines in both stiff and soft contact environments [49] [50]. This demonstrated a highly accurate and **model-free** force sensing approach for telerobotic systems. *Focus: Rigid-body (external force) estimation.*

- **Liu et al. (Robotics & CIM 2021)** – *"Sensorless force estimation for industrial robots using disturbance observer and neural learning of friction approximation."* This work falls under **hybrid (model+learning) rigid-body force estimation**. It combines a classical **disturbance observer** (to estimate generalized forces acting on the robot) with a **neural network** that learns the unmodeled **friction dynamics** [51]. Essentially, the disturbance observer provides a baseline force estimate, and a neural net compensates for the complex nonlinear friction that the observer cannot account for [52]. In experiments on a KUKA industrial robot, the hybrid method reduced friction estimation error by ~66% (friction MSE dropped from 14.67 to 4.97 (Nm)$^2$) and yielded robust end-effector force estimation during various tasks [53] [54]. This showcases the benefit of a residual learning approach: the **neural network handles the hard-to-model part** (friction), improving overall estimation accuracy. *Focus: Rigid-body (force) estimation.*

- **Bao et al. (IEEE T-II 2023)** – *"Adaptive Neural Trajectory Tracking Control for* n*-DOF Robotic Manipulators with State Constraints."* (Represents hybrid control with estimation.) This paper designs a controller that fuses classical and learning elements to handle **rigid-body dynamics uncertainties**. It uses a computed torque controller plus a **radial-basis-function neural network (RBFNN)** to adaptively learn unmodeled dynamics, along with a nonlinear disturbance observer (NDO) and barrier Lyapunov functions for constraint handling [55] [56]. While its focus is on control, it inherently performs online estimation of the uncertainty (via the RBFNN). The result was improved tracking accuracy (lower IAE/ITAE) on a 7-DOF robot compared to traditional controllers [57]. This underscores how hybrid methods can maintain performance under unknown disturbances. *Focus: Rigid-body (dynamics uncertainty estimation in control).*

- **Tao et al. (Robotica 2025)** – *"Robot hybrid inverse dynamics model compensation method based on the BLL residual prediction algorithm."* This recent work is in **hybrid residual learning for rigid-body dynamics**. It introduces a **bagging-ensemble LSTM residual model** (the "BLL" algorithm) that learns to compensate the error between a robot's nominal inverse dynamics model and the true dynamics [58]. By training an LSTM on the torque residuals and aggregating an ensemble of learners, the method significantly improved torque prediction for a Franka Emika manipulator: the average joint torque error dropped from 0.5651 Nm (with the nominal model) to 0.1096 Nm with the residual compensation [59]. This is a **rigid-body estimation** scenario (learning unmodeled effects like flexibility, temperature influence, etc., on the robot's dynamics). The study demonstrates the power of combining physics models with data-driven residual learning to greatly enhance accuracy. *Focus: Rigid-body estimation.*

- **Yang et al. (RCAR 2025)** – *"A Residual-Driven Decomposed PINNs Method for Dynamics Identification of Robot Manipulators."* This conference paper falls under **physics-informed learning for rigid-body**

**dynamics**, potentially including payload. The authors propose a two-stage identification: first use classical least-squares to identify part of the dynamics, then use a **physics-informed neural network (PINN)** to learn the **residual** dynamics (especially nonlinear friction) that the least-squares left unmodeled [60] [61]. The PINN incorporates the known physics (Newton-Euler equations) in its loss function alongside data. On a 6-DOF manipulator, this hybrid PINN approach reduced torque prediction RMSE by up to 65% compared to a standard identification approach [62], showing a robust handling of complex dynamics like friction [63]. *Focus: Rigid-body estimation* (of manipulator dynamics, including friction and potentially payload effects).

- **Zhang et al. (arXiv 2025)** – *"Provably-Safe, Online System Identification."* This preprint addresses **payload estimation** under strict safety constraints. The authors consider a robot lifting an unknown payload and formulate an online identification method that guarantees safety (no joint torque/position limit violations, avoidance of obstacles) throughout the process [64] [65]. They use **interval arithmetic** to keep track of uncertain parameter bounds and an **ARMOUR**-based trajectory optimizer to plan probing motions that are both informative for identification and inherently safe [66] [67]. The result is a provably safe online inertial parameter estimator. Experiments (by simulating a Kinova Gen3 with a dumbbell payload) showed the approach can gradually tighten the payload's parameter bounds (mass, inertia) while never endangering the system [68]. This is a forward-looking combination of identification and safe control. *Focus: Payload estimation.*

- **Taie et al. (IEEE RA-L 2024)** – *"Online Identification of Payload Inertial Parameters Using Ensemble Learning for Collaborative Robots."* This article focuses on **payload estimation** using a modern machine learning approach. The authors develop an **online ensemble learning** method: multiple lightweight models (neural networks) are run in parallel on the robot's proprioceptive data (joint angles, torques), and their outputs are combined to estimate the payload's inertial parameters in real time [69] [70]. The ensemble is updated incrementally as new data comes in. This approach was shown to be both accurate and robust for payload identification during collaborative tasks [71]. It performs well even when individual models face uncertainty, as the ensemble consensus mitigates errors. *Focus: Payload estimation.*

- **Taie et al. (Frontiers in Robotics&AI 2024)** – *"Addressing Catastrophic Forgetting in Payload Parameter Identification Using Incremental Ensemble Learning."* This follow-up work also deals with **payload estimation** and specifically the problem of **knowledge retention** over long-term robot operation. The authors observed that an incremental learning model might "forget" old payload information when adapting to new payloads. They propose a solution: each time a new payload is encountered, a new weak learner is added to the ensemble (rather than overwriting the old model), and a meta-classifier selects the appropriate learner [72] [73]. This preserves knowledge of previously seen payloads. Their results show the method can cycle through different payloads without loss of accuracy – for example, maintaining mass estimate errors on the order of only ~0.007 kg in repeated trials [74]. This innovation allows a collaborative robot to **adapt to varying payloads continually**. *Focus: Payload estimation.*

In summary, the recent state-of-the-art spans a spectrum from classical observers and identification algorithms to hybrid model-learning approaches and purely data-driven methods. Many works target **payload inertial parameter estimation** (reflecting its importance in safe and efficient manipulation), while others focus on **external force/torque estimation** for interaction control. A few, like Hu *et al.* 2025, explicitly tackle **both** robot and payload dynamics. The trend is toward combining physical modeling with machine learning (e.g., Gaussian processes, neural networks, PINNs, ensembles) to improve estimation accuracy and robustness in robotic manipulators [2] [29]. Each "mature" approach above contributes to more reliable and fast estimation of forces or parameters, which is crucial for advanced robotic manipulation in uncertain, human-populated environments. The **focus** of each work – whether on the robot's own dynamics or an attached payload – has been highlighted accordingly. This ensures a clear understanding of how these state-of-the-art methods apply to different aspects of robotic manipulator estimation challenges.

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[1] [2] [2203.00830] Fast Object Inertial Parameter Identification for Collaborative Robots
https://arxiv.org/abs/2203.00830

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https://arxiv.org/abs/2302.06685

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